Software Engineering
UE24CS341A
Lab 07: Fuzz Testing Report

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1. Original Application

The original elevator dispatch application contains three functions to parse elevator requests, calculate travel distance and calculate load percentage. The application works correctly for valid inputs.

```
(.venv) C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7>python elevator_dispatch.py
==== Running parse_request ====
Enter a request as FROM->TO (e.g. 5->9): 5->9
From and To: (5, 9)

==== Running total_travel ====
Enter floors visited as CSV (e.g. 1,5,3,9): 1,5,3,9
Floors Travelled: 12

==== Running load_percent ====
Enter load as 'X of Y' (e.g. 450 of 630): 450 of 630
Load Percent: 71

(.venv) C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7>
```

2. Fuzz Testing

Hypothesis was used to generate random and unexpected inputs for all three functions. Initially, three tests failed and three passed. The failures included empty strings, None values and invalid input formats.

This is a long output from pytest, so multiple screenshots were taken from the same terminal command which ran the pytest

```
(.venv) C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7>python -m pytest test_elevator_dispatch.py -v --tb=short
============================================================ test session starts ============================================================
platform win32 -- Python 3.13.0, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7
plugins: hypothesis-6.170.15
collected 6 items

test_elevator_dispatch.py::test_parse_request_no_crash FAILED                                                                      [ 16%]
test_elevator_dispatch.py::test_total_travel_no_crash FAILED                                                                       [ 33%]
test_elevator_dispatch.py::test_load_percent_no_crash FAILED                                                                       [ 50%]
test_elevator_dispatch.py::test_parse_request_valid PASSED                                                                         [ 66%]
test_elevator_dispatch.py::test_total_travel_valid PASSED                                                                          [ 83%]
test_elevator_dispatch.py::test_load_percent_valid PASSED                                                                          [100%]

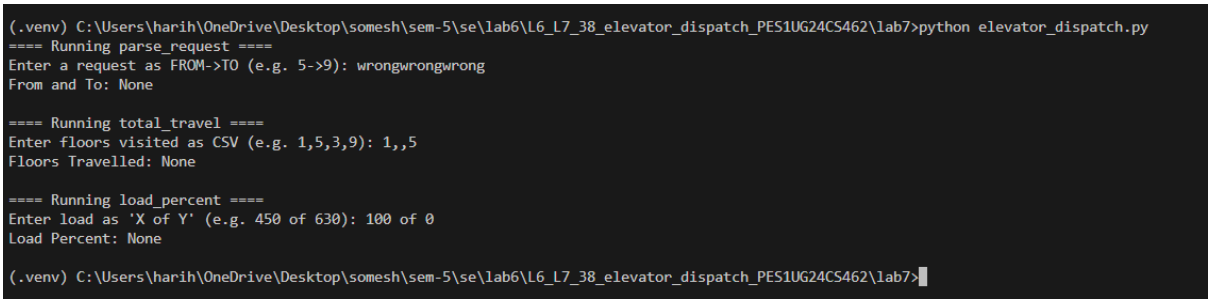
================================================================ FAILURES ================================================================
```

```
=================================================================== FAILURES ===================================================================
_____________________________________________ test_parse_request_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\test_elevator_dispatch.py", line 7, in test_parse_request_no_crash
  |     def test_parse_request_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\test_elevator_dispatch.py", line 8, in test_parse_request_no_crash
    |     result = parse_request(s)
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\elevator_dispatch.py", line 3, in parse_request
    |     return int(start), int(end)
    |            ~~~^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_parse_request_no_crash(
    |     s='->',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\test_elevator_dispatch.py", line 8, in test_parse_request_no_crash
    |     result = parse_request(s)
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\elevator_dispatch.py", line 2, in parse_request
    |     start, end = text.split('->')
    |     ^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_parse_request_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\test_elevator_dispatch.py", line 8, in test_parse_request_no_crash
    |     result = parse_request(s)
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\elevator_dispatch.py", line 2, in parse_request
    |     start, end = text.split('->')
    |                  ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_parse_request_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________________________ test_total_travel_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\test_elevator_dispatch.py", line 13, in test_total_travel_no_crash
  |     def test_total_travel_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\test_elevator_dispatch.py", line 14, in test_total_travel_no_crash
    |     result = total_travel(s)
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\elevator_dispatch.py", line 6, in total_travel
    |     floors = [int(f) for f in floors_csv.split(',')]
    |               ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_total_travel_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\test_elevator_dispatch.py", line 14, in test_total_travel_no_crash
    |     result = total_travel(s)
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\elevator_dispatch.py", line 6, in total_travel
    |     floors = [int(f) for f in floors_csv.split(',')]
    |                              ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_total_travel_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________________________ test_load_percent_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\test_elevator_dispatch.py", line 19, in test_load_percent_no_crash
  |     def test_load_percent_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\test_elevator_dispatch.py", line 20, in test_load_percent_no_crash
    |     result = load_percent(s)
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\elevator_dispatch.py", line 11, in load_percent
    |     return int(load) * 100 // int(capacity)
    |            ~~~^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_load_percent_no_crash(
    |     s=' of ',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\test_elevator_dispatch.py", line 20, in test_load_percent_no_crash
    |     result = load_percent(s)
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\elevator_dispatch.py", line 10, in load_percent
    |     load, capacity = text.split(' of ')
    |     ^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_load_percent_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\test_elevator_dispatch.py", line 20, in test_load_percent_no_crash
    |     result = load_percent(s)
    |   File "C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7\elevator_dispatch.py", line 10, in load_percent
    |     load, capacity = text.split(' of ')
    |                      ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_load_percent_no_crash(
    |     s=None,
    | )
    +------------------------------------
    +------------------------------------
=========================================================== short test summary info ============================================================
FAILED test_elevator_dispatch.py::test_parse_request_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_elevator_dispatch.py::test_total_travel_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_elevator_dispatch.py::test_load_percent_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
========================================================== 3 failed, 3 passed in 3.65s ==========================================================

(.venv) C:\Users\harih\OneDrive\Desktop\somesh\sem-5\se\lab6\L6_L7_38_elevator_dispatch_PES1UG24CS462\lab7>
```

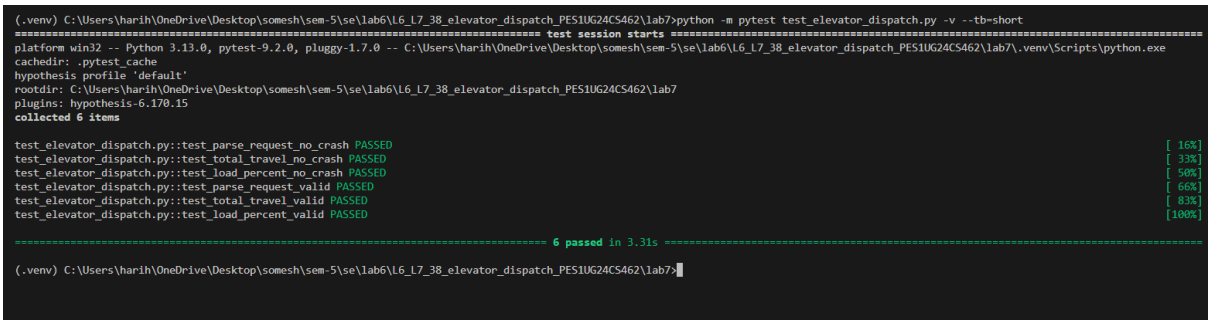
3. Bug Fixes

Input validation was added to all three functions. Invalid inputs are now handled safely without crashing the application. Division by zero is also prevented in the load percentage calculation.



4. Final Testing

The same Hypothesis tests were executed again after fixing the bugs. All six tests passed successfully, showing that the identified input-validation issues were resolved.



5. Conclusion

Fuzz testing helped identify input-validation bugs that normal valid-input testing did not reveal. After the fixes, the application handled the tested invalid inputs without crashing while preserving its normal functionality.